

UCC821: ADVANCED TRANSPORTATION ENGINEERING

L	T	P	Cr
3	1	0	3.5

Course Objectives: To expose students to various road specifications and various aspects of transportation planning and traffic engineering

Road Specifications: Specifications for bitumen roads, bitumen surface treatments, penetration macadam, bitumen bound macadam and bitumen concrete, mastic asphalt, specification for cement concrete roads, construction of bituminous and cement concrete roads.

Traffic control: Traffic signs, Road markings, traffic signals, types i.e. simultaneous system, alternate system, simple progressive system and flexible progressive system, general principles of signal design.

Traffic control aids: Road way delineations, object markers, guardrails, Barriers, Fundamental principles of Traffic Flow, Traffic flow Elements, Flow Density Relationships, Gap and gap acceptance, Introduction of Queing Theory.

Traffic regulations: Traffic laws, Basic principles of regulation, regulation of speed, regulations of vehicles, regulation concerning driver, regulation concerning traffic rules for pedestrian traffic, general rules concerning traffic, enforcement of regulations.

Highway capacity: PCU, Level of service concepts, factors affecting capacity, capacity of urban highways, capacity of rotary intersection.

Intersections: Design of intersection, grade separated intersection, Need for rotary intersection, principles of design, design of rotary intersection.

Transport Planning & Analysis: Basic elements of Transport Planning, Urban Transport System Planning, Level of Urban transport Planning

Evaluating Transportation alternatives: Evaluation Process, Evaluation Criterion, Economic analysis of cost and benefits, Highway Cost, Road User costs, Methods of Economic Analysis

Public Transport System: Purpose & Methods of Public Transport System

Computational Softwares:

Introduction to highway engineering softwares:

1. HEADS
2. MX Road
3. HDM4

Experimental Project/assignment/Micro Project:

1. To design traffic signals for various locations in the city/for a given scenario.
2. To design and propose rotary for any particular location.
3. To perform cost benefit analysis of any highway project.

Course Learning Outcomes (CLO):

Upon completion of this course, the students will be able to:

1. Understand the road specifications laid by Indian road authorities
2. Apply the concepts of traffic engineering including traffic control, highway capacity.
3. Design various intersections.
4. To evaluate the different transportation alternative for a given scenario.

Text Books

1. Khanna S.K. and C.E.G. Justo, "Highway Engineering", Nemchand Bros (2002).
2. Kadyali L. R.; Highway Engineering, Nem Chand & Brothers, Roorkee (2002).
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Reference Books

1. Sharma & Sharma; Principle and Practice of Highway Engineering, Asia Publishing House, New Delhi (1980).
2. Road Development plan for India (1981-2001), IRC, New Delhi (1984).
3. Rao G.V.; Transportation Engineering, Tata McGraw Hill Publisher, New Delhi (1965).
4. Highway Materials, H.M.S.O. (London) (2014).
5. Yoder E.J.; Principles of Pavement Design, John Wiley & Son (1975).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	25